Genome-wide ancestry, climate, and mitotype

A population-level follow-up to the BayPass climate-GEA scan

Corrected version – see Erratum

Generated September 8, 2026

Erratum

An external audit of the first version of this document identified two implementation bugs, both now fixed. All numbers, tables and figures below are corrected; Section also incorporates the audit's remaining recommendations.

1. **Block permutation silently dropped one population per shared-origin pair.** The reduction from 19 populations to 17 independent lineage units was implemented as "take the first population's value, broadcast it to both members of the pair." This was harmless for LångholmenW/LångholmenR (identical climate values) but wrong for Bunkkeri/Grundsund, whose PC1 values differ (-3.409 vs. -3.899) – Grundsund's actual value was discarded, not averaged in. **Fix:** ancestry, climate, and mitotype are now genuinely *averaged* within each shared-origin pair to build a real 17-observation dataset, and every correlation, null, and permutation below operates on that dataset directly.
2. **The Ω-structured null for the partial correlation replaced the wrong variable.** Ω is a nuclear allele-frequency covariance matrix, so it is an appropriate null generator only for the one tested quantity that is itself a direct function of nuclear allele frequencies – ancestry. The original implementation drew random vectors standing in for PC1 (holding ancestry and mitotype fixed) instead of standing in for ancestry (holding PC1 and mitotype fixed). **Fix:** throughout this document, Ω-null draws always replace ancestry. For comparisons where *neither* variable is ancestry (mitotype vs. PC1/PC2/bio6/bio11), the Ω-null is not well-motivated and is omitted; those rows report the block-permutation result only.

Ω itself was reduced to match the 17-unit dataset via $M\Omega M^\top$, where M is the same population-to-lineage-unit averaging operator applied to the phenotypes – propagating the actually-estimated relatedness structure through the reduction, rather than drawing null vectors from an arbitrary 17-dimensional matrix. Corrected results were cross-checked against an independent re-implementation: all r values matched exactly, and p-values agreed closely (small differences are expected Monte Carlo noise between independent permutation runs).

Motivation

The Stage-1-direct BayPass climate-association scan (PC1, PC2, and the winter-temperature variables bio6/bio11) produced raw candidate counts under a descriptive BF(dB)$\geq$15 threshold, but none of the climate axes are expected to survive full structured-null calibration cleanly, because PC1 and PC2 are both partially confounded with genome-wide ancestry (established in `moduleB_ancestry_confound.R`: cor(PC1,ancestry)$= -0.23$, cor(PC2,ancestry)$= +0.57$ across all 20 populations). Rather than treating that confound purely as a caveat that explains away the locus-level scan, this document asks whether the population-level ancestry–climate link is itself real and worth reporting on its own terms – and traces it down to what actually drives it.

All analyses here operate on the population level (*not* individual genotypes), on the `aland_excluded` set of hybrid populations that the BayPass scan itself uses (Åland excluded because its individuals carry mixed mitotypes and it is a known high-leverage outlier in several of the ORIGINAL PC1/PC2 diagnostics), further reduced to **17 independent lineage units** as described in the Erratum.

Two non-independence-aware significance tests

A plain Pearson correlation across populations overstates the effective sample size: two population pairs have a documented shared origin from the phylogenetic network analysis (LångholmenW+LångholmenR; Bunkkeri+Grundsund; Nouhaud et al. 2022), and all populations share allele-frequency covariance captured by the fixed population covariance matrix Ω already estimated for the BayPass scan (from all 698,251 Stage-1 cluster representatives). Two complementary null designs are used throughout, both now operating on the genuine 17-lineage-unit dataset:

- **Ω-structured null.** 10,000 random vectors are drawn from the 17-unit-reduced Ω's eigendecomposition, standing in for ancestry (see Erratum). Empirical $p = (1 + k)/(n_{\text{sim}} + 1)$, $k =$ number of null draws with $|r_{\text{null}}| \geq |r_{\text{obs}}|$.
- **Shared-origin block permutation.** 10,000 permutations shuffle the tested variable across the 17 independent units directly, recomputing the correlation each time.

Benjamini-Hochberg correction is applied within each family of tests (the 4 climate variables; the 5 mitotype comparisons), using the block-permutation p-values as the primary statistic, since the Ω-null is not defined for every row.

1. Ancestry vs. climate axes

Population ancestry (mean oriented *F. aquilonia* allele frequency at diagnostic markers, DI> -25, $|\Delta p| \geq 0.5$) is compared against the two original climate axes (PC1, PC2) and the two new winter-temperature variables suggested for the GEA scan (bio6 = min. temperature of the coldest month; bio11 = mean temperature of the coldest quarter).

Variable	r (17 units)	Ω-null p	Block-perm. p	Block-perm. p_{BH}
PC1	-0.481	0.044	0.051	0.164
PC2	$+0.381$	0.110	0.123	0.164
bio6	$+0.404$	0.091	0.110	0.164
bio11	$+0.287$	0.210	0.267	0.267

Table 1: Ancestry vs. climate, 17 independent lineage units. p_{BH} = Benjamini-Hochberg corrected across the 4 tests.

Finding. PC1 has the strongest ancestry association among the four variables and is the only one to fall below $p = 0.05$ under the block-permutation test ($p = 0.051$, i.e. right at the boundary) – but the two null designs disagree (Ω-null $p = 0.044$ vs. block-permutation $p = 0.051$), and **none of the four variables survive Benjamini-Hochberg correction** (minimum $p_{\text{BH}} = 0.164$). The accurate summary is that PC1's ancestry association is *suggestive but not formally significant*, not that it "clears" a defensible statistical test. bio6 is the next-closest candidate but is weaker on both nulls than PC1.

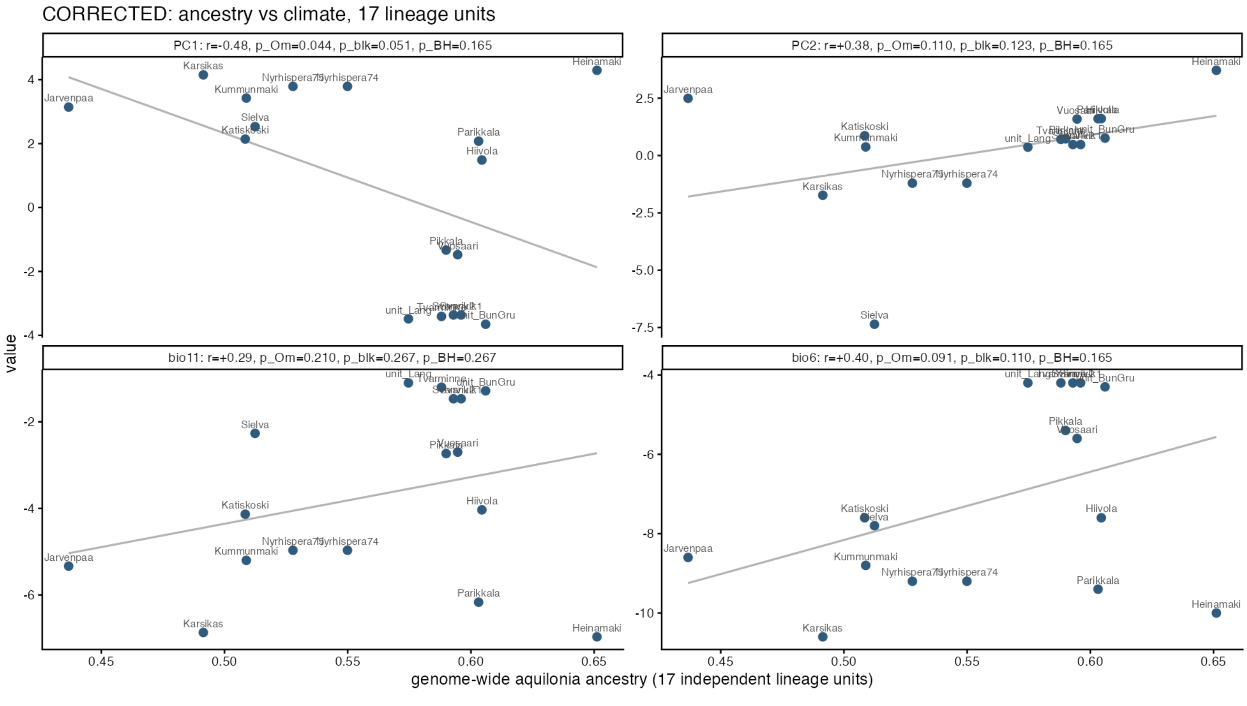

Figure 1: Ancestry vs. PC1/PC2/bio6/bio11, 17 independent lineage units, with corrected statistics annotated per panel.

2. What drives the PC1–ancestry signal?

Three complementary diagnostics identify which population(s) are responsible for the PC1–ancestry correlation: leave-one-population-out, leave-one-independent-unit-out, and standard regression influence statistics (Cook's distance, standardized residuals) from lm(ancestry ∼ PC1). *This diagnostic predates the 17-unit correction and is reported on the original 19-population scale* (observed $r = -0.502$) – Heinamäki and Jarvenpaa are not part of either shared-origin pair, so the bug did not affect their values, and the qualitative finding is unchanged at 17-unit resolution (raw $r = -0.481$; see Section 1).

Population	Leave-one-out Δr	r without it	Cook's D	Std. residual
Heinamäki	−0.211	−0.712	**0.618**	+2.75
Jarvenpaa	+0.030	−0.471	0.288	−2.30
all others	$\lvert\leq 0.06\rvert$	–	≤ 0.07	$\lvert\leq 1.2\rvert$

Table 2: Regression influence diagnostics for PC1 vs. ancestry, 19-population scale (observed $r = -0.502$). Rule-of-thumb cutoffs: Cook's $D > 4/n = 0.211$.

Finding. Heinamäki dominates every diagnostic by a wide margin, with Jarvenpaa a distant second; every other population's individual influence is negligible. Critically, **Heinamäki dampens rather than drives** the correlation – it combines high ancestry (0.651) with an extreme PC1 value (4.29) in a way that sits *off* the negative trend, so excluding it makes the correlation *stronger* ($r: -0.502 \to -0.712$). Whatever one makes of PC1's borderline significance (Section 1), it is not an artifact of one population manufacturing a spurious signal.

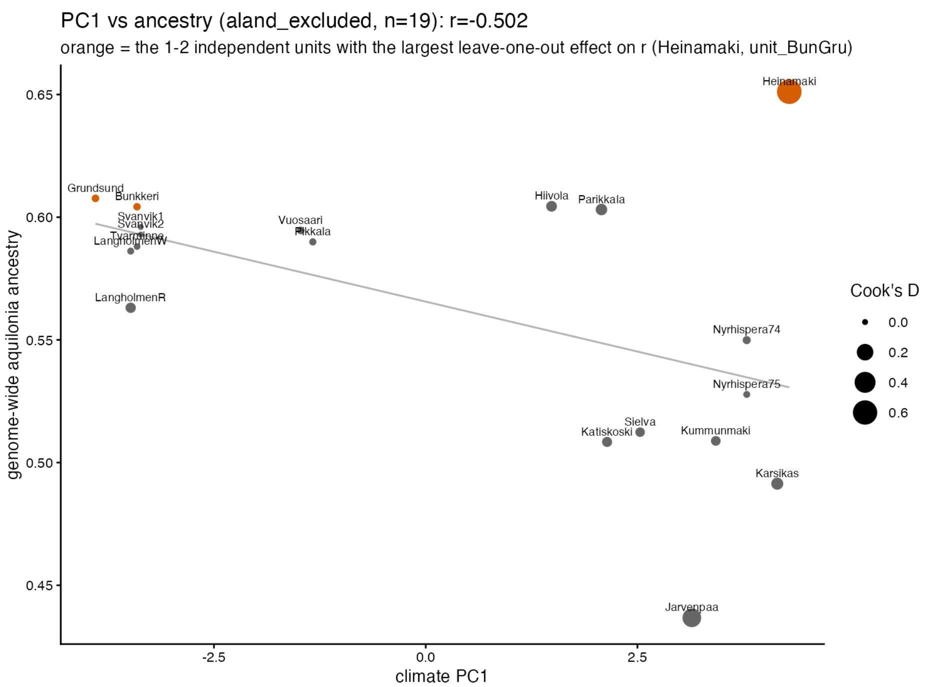

Figure 2: PC1 vs. ancestry (19-population scale), point size = Cook's distance. Heinamäki (largest point) is the dominant influence but pulls *against* the fitted trend.

Is Heinamäki itself anomalous?

Three checks on Heinamäki's own data found nothing unusual: mitotype is uniform (10/10 *F. aquilonia*-like, not mixed like Åland); it was sampled from 2 nests (159, 160), typical of the panel; and its hybrid index (0.357 ± 0.017) and heterozygosity (0.314 ± 0.038) sit comfortably among the other *F. aquilonia*-skewed mainland populations. Admixture timing (DATES) could not be checked here (not part of this repository).

3. Mitotype vs. ancestry and climate

Mitotype (*F. aquilonia*-like vs. *F. polyctena*-like maternal lineage) is uniform within both shared-origin pairs, so it survives the 17-unit reduction unambiguously (6 *F. aquilonia*-like, 11 *F. polyctena*-like units). Tested as a population-level binary variable (point-biserial correlation) against the same five quantities.

Variable	r with mitotype	Ω-null p	Block-perm. p	Block-perm. p_{BH}
ancestry	+0.490	0.037	0.049	0.243
PC1	−0.299	–	0.255	0.424
PC2	+0.197	–	0.508	0.508
bio6	+0.291	–	0.250	0.424
bio11	+0.219	–	0.396	0.494

Table 3: Mitotype vs. ancestry/climate, 17 units. Ω-null omitted where neither variable is ancestry (see Erratum). p_{BH} = Benjamini-Hochberg corrected across the 5 tests.

Finding. Mitotype and genome-wide nuclear ancestry are nominally associated ($p \approx 0.037$–0.049 uncorrected) but **this does not survive correction for the 5-comparison family** ($p_{\mathrm{BH}} = 0.243$). No climate variable shows any evidence of a mitotype association. The most defensible description is an *association between mitochondrial lineage and genome-wide nuclear ancestry, consistent with shared demographic history* – language such as "founder-history signal" or "populations founded by an *F. aquilonia*-like queen" asserts a specific causal mechanism (founding history) that this correlational analysis cannot distinguish from alternatives such as asymmetric introgression, retained population structure, or selection involving mitochondrial ancestry.

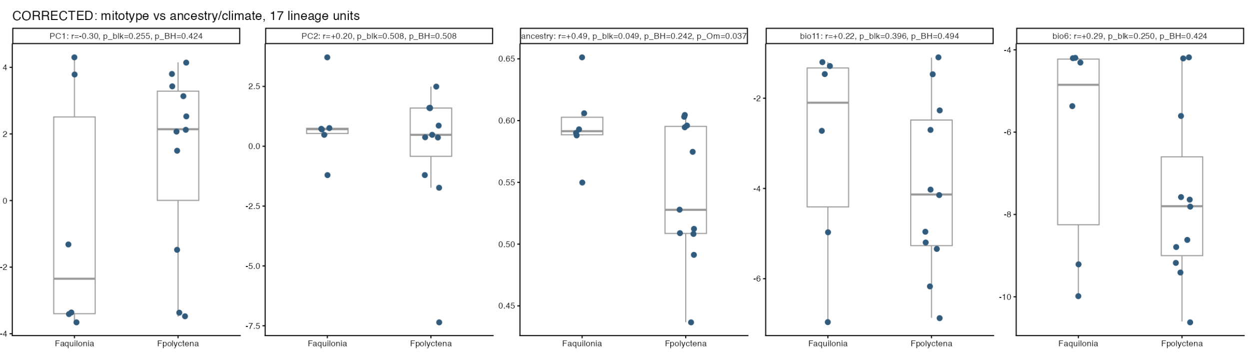

Figure 3: Mitotype vs. each variable, 17 independent lineage units.

4. Does PC1's ancestry link survive controlling for mitotype?

Since mitotype correlates with both ancestry and, more weakly, PC1, the PC1–ancestry correlation could partly reflect the same demographic-history structure linking mitotype to ancestry rather than an independent signal. Tested via partial correlation on the 17-unit dataset, with both nulls re-run on the partial statistic itself (Ω-null draws correctly replacing ancestry; see Erratum).

	r (17 units)	Notes
Raw cor(PC1, ancestry)	−0.481	Ω-null $p = 0.044$; block-perm. $p = 0.051$
Partial cor(PC1, ancestry | mitotype)	−0.402	Ω-null $p = 0.097$; block-perm. $p = 0.124$

Table 4: PC1–ancestry correlation before/after controlling for mitotype, 17 units.

Finding. Controlling for mitotype moderately attenuates the PC1–ancestry correlation (−0.481 → −0.402), and the remaining association is not distinguishable from the population-structure null under either test ($p = 0.097$–0.124). This is consistent with two readings – a genuinely independent, if weaker, climate signal; or partial confounding by mitotype-linked demographic history – and **the data cannot distinguish between them**. With only 17 independent lineage units, this sample cannot separate the contributions of climate and maternal-lineage history with any confidence; that is a limitation of the design, not a result in either direction.

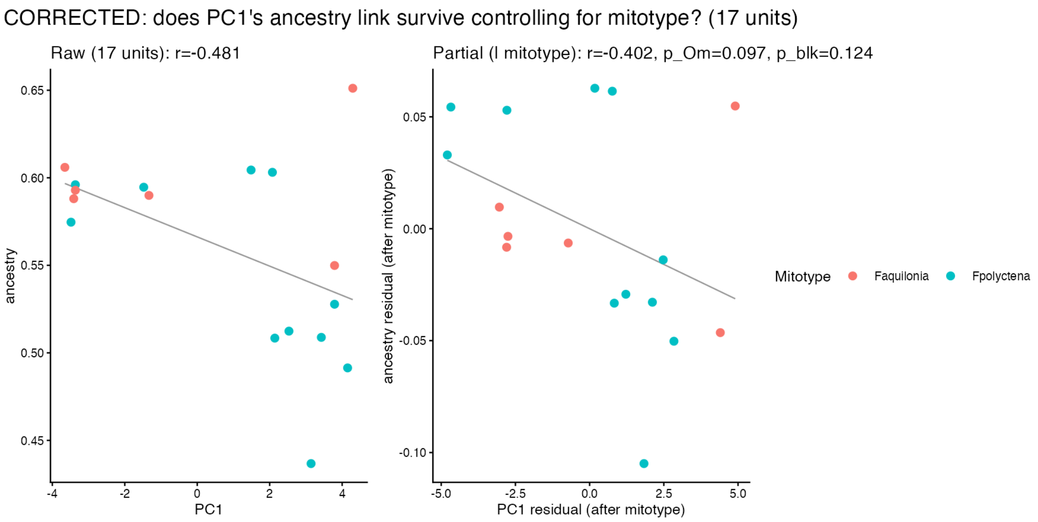

Figure 4: PC1 vs. ancestry, raw (left) and after removing the linear effect of mitotype from both variables (right), 17 units.

5. Relationship to the BayPass locus-level scan

The population-level tests above and the BayPass climate-association scan ask related but *different* questions. The population-level test asks whether a climate variable correlates with *one genome-wide ancestry summary* across populations. BayPass asks whether *particular loci or LD clusters* are unusually associated with that climate variable, after accounting for population covariance (Ω) and thousands of tests. A population-level correlation being weak or non-significant does not, by itself, resolve whether it is relevant to the locus-level scan – the two statements below need to be held together.

The climate axes are moderately aligned with population structure, but that alignment is not demonstrably stronger than expected by chance under that structure. For PC1, the ancestry correlation is moderately large after excluding Åland ($r \approx -0.48$ on the corrected 17-unit dataset), but it is only suggestive under the Ω-null ($p = 0.044$), nominal under the corrected lineage permutation ($p \approx 0.051$), and non-significant after controlling for mitotype ($p = 0.097$–0.124; Section 4). PC2 is weaker after Åland exclusion and clearly non-significant under both structure-aware tests (Section 1).

This is fully compatible with, and plausibly explains, both the original canonical BayPass climate scan's null-calibrated result and the new Stage-1-direct scan's. A moderate alignment between climate and ancestry can cause loci that strongly differentiate the underlying population groups to acquire large *raw* Bayes factors, simply because those loci track the same population split the climate variable happens to track. But Ω-structured null covariates – random vectors sharing only the populations' covariance structure, with no causal link to climate at all – frequently produce similarly-aligned, similarly-sized apparent outliers. Consequently the observed candidate counts were not exceptional against that null, under *either* scan:

	Stage-2 eMLG scan (32,854 clusters)	Stage-1-direct scan (18,361 units)	Expected by
PC1 floor survivors	0	3	1.84 (Stage-1) / 3.29 (S
PC2 floor survivors	4	3	1.84 (Stage-1) / 3.29 (S
Set-level FDR	≈ 0.82 (PC2)	≈ 0.61 (both axes)	

Table 5: Floor-survivor candidate counts (BF(dB)$\geq$15 and beating all 10,000 Ω-structured null covariates) under both BayPass scans. "Expected by chance" is $N_{\text{tested}}/(n_{\text{sim}} + 1)$.

Both scans land at a high set-level FDR (≥ 0.6) – neither is a defensible discovery set. The one notable shift is that PC1, which had *zero* floor survivors under the more heavily Stage-2-merged scan, has 3 under the less-diluted Stage-1-direct scan (BF(dB) = 53.7, 38.7, 31.1) – consistent with, though not confirming, the population-level observation above that PC1 (not PC2) is the axis with the more suggestive ancestry link. PC2's candidate count is essentially unchanged (4 vs. 3) and remains similarly unremarkable relative to its own null expectation. The structured null is, in effect, saying: *these apparent climate associations are approximately what is expected when a covariate happens to align with population history, regardless of how the markers were reduced to test units.*

Confounding does not have to be statistically significant to influence locus-level results. Statistical significance asks whether the observed alignment is unusually strong given uncertainty and structure; confounding merely describes that climate and ancestry point partly in the same direction. With only 17–19 independent populations, a correlation of 0.4–0.5 can be biologically noticeable, and large enough to shift which loci look like outliers in a raw BayPass scan, while still being far too weak for a population-level test with this little power to call it exceptional.

Bottom line. Both climate axes were partially aligned with genome-wide ancestry, although these population-level correlations were not consistently distinguishable from expectations under the population-covariance null. This moderate alignment nevertheless provides a plausible source of the raw locus-level BayPass signal: loci that differentiate the same population groups can appear climate-associated even without any locus-specific environmental effect. Consistent with this interpretation, neither the number nor the strength of cluster-level BayPass associations exceeded what was obtained under Ω-structured null covariates, under *either* the Stage-2 or the Stage-1-direct scan (set-level FDR ≥ 0.6 in both). The conclusion is *not* that climate definitely has no relationship with ancestry – it is that the data cannot separate a climate effect from population history, and the BayPass outliers provide no additional locus-specific evidence beyond that structure. Even a formally significant genome-wide PC1–ancestry correlation would not, on its own, rescue the BayPass candidates: it would show that overall ancestry covaries with PC1, but evidence for particular climate-associated genomic regions would still require their Bayes factors to exceed the structured-null distribution, which they did not.

Limitations and recommended follow-up

This analysis should be treated as **supplementary and explicitly exploratory**, not a confirmed result, for several reasons:

- **Only PC1 received the full mitotype-controlled (partial) analysis.** PC2, bio6, and bio11 are examined only through bivariate correlations. The conclusion that no climate variable shows an ancestry- and mitotype-independent signal is broader than what has actually been tested for those three variables; either the same partial model should be run for all four, or the conclusion should be narrowed to PC1 specifically.
- **Ancestry was computed once, from the frozen DI> -25 marker panel.** It has not been recomputed from an independent marker set or from LD-reduced diagnostic units, so its sensitivity to the specific marker panel used is untested.
- **A single coherent structure-aware model is preferable to a sequence of bivariate/partial tests.** A model of the form ancestry $\sim$ climate + mitotype with an Ω-based residual covariance (or an equivalent parametric bootstrap) would estimate both predictors' contributions jointly, with correctly propagated uncertainty, rather than inferring their relative roles from separate bivariate/partial tests as done here.
- With 17 independent lineage units and 4-5 tests per family, this analysis is underpowered to detect anything but a fairly strong effect, and Ω-null and block-permutation p-values disagree by up to $\sim$0.03–0.05 in places (e.g. PC1: 0.044 vs. 0.051) – a reminder that "significant" here would depend on which defensible null is treated as primary, not a settled question either null can resolve alone.

Summary

Across the 19 Åland-excluded populations (17 independent lineage units after correcting for shared origins), genome-wide ancestry showed a moderate negative correlation with climate PC1 ($r = -0.48$). This relationship was nominally suggestive under the shared-lineage permutation ($p = 0.051$) but did not clear the Ω-structured null ($p = 0.044$ is itself only marginal, and the two nulls disagree), and it did not survive correction for testing four climate variables ($p_{\text{BH}} = 0.16$). The association was further attenuated and statistically indistinguishable from the population-structure null after controlling for mitotype ($r = -0.40$, $p = 0.10$–0.12). Mitotype was itself nominally associated with nuclear ancestry ($p \approx 0.04$) but not with any measured climate variable, and that association also did not survive multiple-testing correction. These patterns suggest contributions from population history and possibly climate, but the limited number of independent populations does not permit their effects to be separated confidently. Critically, this moderate, inconclusive population-level alignment is sufficient to explain the raw candidate counts in the original BayPass climate scan without invoking any locus-specific climate effect (Section 5): no PC1 cluster and only a null-expected number of PC2 clusters exceeded the Ω-structured null. The BayPass scan therefore provides no locus-specific evidence for climate adaptation beyond what population structure already predicts. This document should be read as a supplementary, exploratory analysis, not a confirmed finding for the main text.